PH3120 – Computational Physics Laboratory 1 CPL104 – Vectors and Matrices Section 1 & 2

1.Vectors

I).

(a)

Scalar Quantity	Vector Quantity
Has only magnitude	Has both magnitude and direction
No direction	Always has a direction
Represented by a number and unit	Represented by an arrow (magnitude + direction)
Examples : Speed, Mass, Temperature, Time, Energy	Examples :Velocity, Displacement, Force, Acceleration

(b)

$$\vec{V} = Xi + Yj + Kz$$

Magnitude of $\vec{V}$:

$$|\vec{V}| = \sqrt{X^2 + Y^2 + K^2}$$

(c) if $\vec{A}, \vec{B}$ are vectors ,

$$\vec{A}.\vec{B} = |a||b|cos\theta$$

a= Magnitude value of $\vec{A}$

b= Magnitude value of $\vec{B}$

d)

$\vec{A}, \vec{B}$ are vectors ,

$$\vec{A} \times \vec{B} = |a||b|sin\theta\ \tilde{n}$$

OR

$\vec{A}$= **(**A_1, A_2, A_3)

$\vec{B}$= (B_1, B_2, B_3)

Then the cross product is:

$$\vec{A} \times \vec{B} = \begin{vmatrix} i & j & k \\ A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \end{vmatrix} = i(A_2B_3 - A_3B_2) - j(A_1B_3 - A_3B_1) + k(A_1B_2 - A_2B_1)$$

e)

$\vec{A}.\vec{B} = |a||b|cos\theta$

$\theta = cos^{-1}\{\frac{\vec{A}.\vec{B}}{|a||b|}\}$

f)

$$unit\ vector\ of\ \vec{V} = \frac{\vec{V}}{|\vec{V}|}$$

II).

a)

[2]:
```
# Q 1)
```

[27]:
```
import numpy as np
k = 8.988e9  # 1 / (4 * pi * epsilon)
charges = np.array([2e-6, -4e-6, 6e-6])
positions = np.array([[1, 0],[0, 2], [-2, -1]])
Electric_filed = np.array([0.0, 0.0])
print("Electric fileds :")
for i in range (0,3):
    q = charges[i]
    r = np.linalg.norm(positions[i])  # distance form charge to origin or magnitude of vect
    vector = -positions[i]
    unit_vector= vector/r
    E= ((k*q)/(r**2))*unit_vector
    print (E)
    Electric_filed += E
print("\n")
print("Total Electric filed of origin : \n",Electric_filed)
print("\n")
magnitude = np.linalg.norm(Electric_filed)
angle = np.degrees(np.arctan2(Electric_filed[1], Electric_filed[0]))  # angle from x-axis

print(" Magnitude of Total Electric Field :\n ",magnitude)
print("\n")
print(" Direction (angle from x-axis in degrees):\n",angle)
print("\n")
unit_vector = Electric_filed / magnitude
print("Unit Vector of Total Electric Field:\n", unit_vector)
```

```
Electric fileds :
[-17976.      0.]
[  -0. 8988.]
[9646.93391125 4823.46695562]


Total Electric filed of origin :
 [-8329.06608875 13811.46695562]


 Magnitude of Total Electric Field :
  16128.544924359738


 Direction (angle from x-axis in degrees):
 121.09226324983518


Unit Vector of Total Electric Field:
 [-0.5164177   0.85633683]
```

b)

```
8]: # Q 1) (b)
```

```
0]: import numpy as np

    # Moment of inertia tensor (3x3 matrix)
    I = np.array([
        [3, 0, 0],
        [0, 5, 0],
        [0, 0, 4]
    ])


    w = np.array([0, 0, 5])


    L = I*w


    print("Angular Momentum Vector (kg·m²/s):", L)
```

```
Angular Momentum Vector (kg·m²/s): [[ 0  0  0]
 [ 0  0  0]
 [ 0  0 20]]
```

c)

```
]: # Q 1) (c)

1]: import numpy as np

q = 2e-6
v = np.array([3, 4, 0])
B = np.array([0, 0.2, 0.3])

F = q * np.cross(v, B)

print("Magnetic Force Vector (N):", F)

Magnetic Force Vector (N): [ 2.4e-06 -1.8e-06  1.2e-06]
```

2)

I. (a)

matrix addition	matrix multiplication
Commutative A+B = B+A	Associative Property A(BC)=(AB)C
Associative (A+B)+C=A+(B+C)	Distributive Property A(B+C) = AB + AC
Additive Identity A+0 = A	Multiplicative Identity AI=IA= A
Additive Inverse A + (-A) = 0	Not Commutative AB≠BA

(b)

Matrix transposition (A^T) = It means flipping a matrix over its diagonal. Rows become columns, and columns become rows.

Effect on Dimensions - If matrix A is m × n, then A^T becomes n × m.

(c)

The inverse of a matrix A, written as A^{-1}, is a matrix that "undoes" the effect of A.
It satisfies:

$$A.\ A^{-1} = A^{-1}.A=I$$

Where III is the identity matrix.

Conditions to have an inverse:

- Should be a square matrix.
- It’s determinant is not be zero.

(d).

A square matrix has the same number of rows and columns**.**

But a rectangular matrix has a different number of rows and columns. And only square matrices can have properties like determinant or inverse.

(e).

A diagonal matrix is a square matrix where all off-diagonal elements are zero**.**
Only the numbers on the main diagonal (top-left to bottom-right) can be non-zero.

Ex :

$$\begin{bmatrix} 6 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 10 \end{bmatrix}$$

(f)

An identity matrix is a square matrix with:

- 1's on the main diagonal
- 0's everywhere else

It’s like the number 1 for matrices.

(g)

The rank of a matrix is the number of linearly independent rows or columns.

It tells:

- How much useful **(**non-redundant**)** information is in the matrix.
- The dimension of the space spanned by its rows or columns.

Relationship to Linear Independence:

- If columns (or rows) are linearly independent, they contribute to the rank.
- If a column is a combination of others, it does not increase the rank.

(h)

Yes, the rank of a matrix can change depending on the operation.

Addition – can be change. Depends on how matrices cancel out.

Multiplication – Can be change. Rank can decrease

Inversion – Can't be change. Only invertible (full-rank) matrices

3)

d) Given a square matrix A, a non-zero vector v is called an eigenvector if:

$$Av=\lambda v$$

Here:

- λ is a scalar called the eigenvalue
- v is the eigenvector
- This means: multiplying A by v just stretches or shrinks v, without changing its direction.

e)

The characteristic equation is a special equation used to find the eigenvalues of a square matrix.

To find eigenvalues:

1. Start with:

   $$Av=\lambda v$$

   Rewriting:

   $$(A-\lambda I)v=0$$

2. For a non-zero v to exist, the matrix $(A-\lambda I)$ must be singular (not invertible), so:

   $$\det(A-\lambda I)=0$$

3. This is called the characteristic equation. Solving it gives the eigenvalues λ

f).

- Matrix transformation changes a vector's direction and length.
- Eigenvectors are special: when transformed, their direction doesn't change, only their length is scaled by the eigenvalue.

If A represents a transformation (like stretching or rotating):

- Eigenvectors = directions that stay the same
- Eigenvalues = how much those directions are stretched or shrunk

4)

I. a)

A system of linear equations can be written as:

b)

For a square coefficient matrix A (same number of equations as variables), the determinant det (A) tells us:

- If $\det(A) \neq 0$, then A is invertible, and the system has a unique solution.
- If $\det(A)=0$, the matrix is singular, and the system may have no solution or infinitely many solutions (depending on consistency).

c)

If A is invertible ($\det(A) \neq 0$), then:

$Ax = b \implies x = A^{-1}b$

where A^{-1} is the inverse of matrix A.